

Table creation :

```
-- CREATE database CompanyDB;
```

```
-- USE CompanyDB;
```

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary INT,  
    hire_date DATE,  
    city VARCHAR(50)  
);
```

```
INSERT INTO Employees (emp_id, emp_name, dept_id, salary, hire_date, city)  
VALUES
```

```
(101, 'Anuj', 1, 50000, '2022-01-10', 'Delhi'),  
(102, 'Ravi', 2, 60000, '2021-03-15', 'Mumbai'),  
(103, 'Neha', 1, 55000, '2020-07-20', 'Delhi'),  
(104, 'Amit', 3, 70000, '2019-11-01', 'Pune'),  
(105, 'Sneha', 2, 65000, '2023-05-12', 'Mumbai'),  
(106, 'Karan', 3, 72000, '2018-08-25', 'Bangalore'),  
(107, 'Pooja', 1, 48000, '2022-09-30', 'Delhi'),  
(108, 'Raj', 2, 52000, '2021-12-11', 'Chennai'),  
(109, 'Meena', 3, 68000, '2020-02-14', 'Pune'),  
(110, 'Vikas', 1, 51000, '2023-01-18', 'Delhi');
```

```
CREATE TABLE Departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
INSERT INTO Departments (dept_id, dept_name) VALUES
```

```
(1, 'HR'),  
(2, 'Sales'),  
(3, 'IT');
```

```
CREATE TABLE Projects (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(50),  
    dept_id INT,  
    budget INT  
);
```

```
INSERT INTO Projects (project_id, project_name, dept_id, budget) VALUES
(201, 'Alpha', 1, 100000),
(202, 'Beta', 2, 150000),
(203, 'Gamma', 3, 200000),
(204, 'Delta', 2, 120000);
```

```
CREATE TABLE Attendance (
    emp_id INT,
    date DATE,
    status VARCHAR(10)
);
```

```
INSERT INTO Attendance (emp_id, date, status) VALUES
(101, '2024-03-01', 'Present'),
(102, '2024-03-01', 'Absent'),
(103, '2024-03-01', 'Present'),
(104, '2024-03-01', 'Present'),
(105, '2024-03-01', 'Absent'),
(106, '2024-03-01', 'Present'),
(107, '2024-03-01', 'Present'),
(108, '2024-03-01', 'Present'),
(109, '2024-03-01', 'Absent'),
(110, '2024-03-01', 'Present');
```

Solutions:

Answer 1:

SELECT * FROM Employees;

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	102	Ravi	2	60000	2021-03-15	Mumbai
	103	Neha	1	55000	2020-07-20	Delhi
	104	Amit	3	70000	2019-11-01	Pune
	105	Sneha	2	65000	2023-05-12	Mumbai
	106	Karan	3	72000	2018-08-25	Bangalore
	107	Pooja	1	48000	2022-09-30	Delhi
	108	Raj	2	52000	2021-12-11	Chennai
	109	Meena	3	68000	2020-02-14	Pune
	110	Vikas	1	51000	2023-01-18	Delhi
★	NULL	NULL	NULL	NULL	NULL	NULL

Answer 2:

SELECT emp_name, salary FROM Employees;

	emp_name	salary
▶	Anuj	50000
	Ravi	60000
	Neha	55000
	Amit	70000
	Sneha	65000
	Karan	72000
	Pooja	48000
	Raj	52000
	Meena	68000
	Vikas	51000

Answer 3:

SELECT * FROM Employees WHERE dept_id = 1;

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	103	Neha	1	55000	2020-07-20	Delhi
	107	Pooja	1	48000	2022-09-30	Delhi
	110	Vikas	1	51000	2023-01-18	Delhi
*	NULL	NULL	NULL	NULL	NULL	NULL

Answer 4:

SELECT * FROM Employees WHERE city = 'Delhi';

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	103	Neha	1	55000	2020-07-20	Delhi
	107	Pooja	1	48000	2022-09-30	Delhi
	110	Vikas	1	51000	2023-01-18	Delhi
*	NULL	NULL	NULL	NULL	NULL	NULL

Answer 5:

SELECT * FROM Employees WHERE salary > 60000

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	104	Amit	3	70000	2019-11-01	Pune
	105	Sneha	2	65000	2023-05-12	Mumbai
	106	Karan	3	72000	2018-08-25	Bangalore
	109	Meena	3	68000	2020-02-14	Pune
*	NULL	NULL	NULL	NULL	NULL	NULL

Answer 6:

SELECT * FROM Employees WHERE hire_date > '2021-12-31';

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	105	Sneha	2	65000	2023-05-12	Mumbai
	107	Pooja	1	48000	2022-09-30	Delhi
	110	Vikas	1	51000	2023-01-18	Delhi
*	NULL	NULL	NULL	NULL	NULL	NULL

Answer 7:

SELECT DISTINCT city FROM Employees;

	city
▶	Delhi
	Mumbai
	Pune
	Bangalore
	Chennai

Answer 8:

SELECT * FROM Employees ORDER BY salary DESC;

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	106	Karan	3	72000	2018-08-25	Bangalore
	104	Amit	3	70000	2019-11-01	Pune
	109	Meena	3	68000	2020-02-14	Pune
	105	Sneha	2	65000	2023-05-12	Mumbai
	102	Ravi	2	60000	2021-03-15	Mumbai
	103	Neha	1	55000	2020-07-20	Delhi
	108	Raj	2	52000	2021-12-11	Chennai
	110	Vikas	1	51000	2023-01-18	Delhi
	101	Anuj	1	50000	2022-01-10	Delhi
	107	Pooja	1	48000	2022-09-30	Delhi
*	NULL	NULL	NULL	NULL	NULL	NULL

Answer 9:

```
SELECT COUNT(*) AS Total_Employees FROM Employees;
```

	Total_Employees
▶	10

Answer 10:

```
SELECT AVG(salary) AS Average_Salary FROM Employees;
```

	Average_Salary
▶	59100.0000

Answer 11:

```
SELECT dept_id, SUM(salary) AS Total_Salary FROM Employees GROUP BY dept_id;
```

	dept_id	Total_Salary
▶	1	204000
	2	177000
	3	210000

Answer 12:

```
SELECT dept_id, COUNT(*) AS Employee_Count FROM Employees GROUP BY dept_id;
```

	dept_id	Employee_Count
▶	1	4
	2	3
	3	3

Answer 13:

```
SELECT dept_id, MAX(salary) AS Max_Salary FROM Employees GROUP BY dept_id;
```

	dept_id	Max_Salary
▶	1	55000
	2	65000
	3	72000

Answer 14:

```
SELECT e.emp_name, d.dept_name  
FROM Employees e  
JOIN Departments d ON e.dept_id = d.dept_id;
```

	emp_name	dept_name
▶	Anuj	HR
	Ravi	Sales
	Neha	HR
	Amit	IT
	Sneha	Sales
	Karan	IT
	Pooja	HR
	Raj	Sales
	Meena	IT
	Vikas	HR

Answer 15:

```
SELECT e.* FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Sales';
```

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	102	Ravi	2	60000	2021-03-15	Mumbai
	105	Sneha	2	65000	2023-05-12	Mumbai
	108	Raj	2	52000	2021-12-11	Chennai

Answer 16:

```
SELECT * FROM Employees WHERE emp_name LIKE 'A%';
```

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	104	Amit	3	70000	2019-11-01	Pune
•	NULL	NULL	NULL	NULL	NULL	NULL

Answer 17:

```
SELECT * FROM Employees WHERE salary BETWEEN 50000 AND 70000;
```

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	102	Ravi	2	60000	2021-03-15	Mumbai
	103	Neha	1	55000	2020-07-20	Delhi
	104	Amit	3	70000	2019-11-01	Pune
	105	Sneha	2	65000	2023-05-12	Mumbai
	108	Raj	2	52000	2021-12-11	Chennai
	109	Meena	3	68000	2020-02-14	Pune
	110	Vikas	1	51000	2023-01-18	Delhi
•	NULL	NULL	NULL	NULL	NULL	NULL

Answer 18:

SELECT * FROM Employees WHERE city <> 'Mumbai';

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	103	Neha	1	55000	2020-07-20	Delhi
	104	Amit	3	70000	2019-11-01	Pune
	106	Karan	3	72000	2018-08-25	Bangalore
	107	Pooja	1	48000	2022-09-30	Delhi
	108	Raj	2	52000	2021-12-11	Chennai
	109	Meena	3	68000	2020-02-14	Pune
	110	Vikas	1	51000	2023-01-18	Delhi
●	NULL	NULL	NULL	NULL	NULL	NULL

Answer 19:

SELECT * FROM Employees WHERE hire_date LIKE '2022%';

-- OR: WHERE YEAR(hire_date) = 2022;

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	101	Anuj	1	50000	2022-01-10	Delhi
	107	Pooja	1	48000	2022-09-30	Delhi
●	NULL	NULL	NULL	NULL	NULL	NULL

Answer 20:

SELECT * FROM Employees ORDER BY salary DESC LIMIT 3;

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	106	Karan	3	72000	2018-08-25	Bangalore
	104	Amit	3	70000	2019-11-01	Pune
	109	Meena	3	68000	2020-02-14	Pune
●	NULL	NULL	NULL	NULL	NULL	NULL

Answer 21:

SELECT * FROM Employees

INNER JOIN Departments ON Employees.dept_id = Departments.dept_id;

	emp_id	emp_name	dept_id	salary	hire_date	city	dept_id	dept_name
►	101	Anuj	1	50000	2022-01-10	Delhi	1	HR
	102	Ravi	2	60000	2021-03-15	Mumbai	2	Sales
	103	Neha	1	55000	2020-07-20	Delhi	1	HR
	104	Amit	3	70000	2019-11-01	Pune	3	IT
	105	Sneha	2	65000	2023-05-12	Mumbai	2	Sales
	106	Karan	3	72000	2018-08-25	Bangalore	3	IT
	107	Pooja	1	48000	2022-09-30	Delhi	1	HR
	108	Raj	2	52000	2021-12-11	Chennai	2	Sales
	109	Meena	3	68000	2020-02-14	Pune	3	IT
	110	Vikas	1	51000	2023-01-18	Delhi	1	HR

Answer 22:

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d ON e.dept_id = d.dept_id;
```

	emp_name	dept_name
►	Anuj	HR
	Ravi	Sales
	Neha	HR
	Amit	IT
	Sneha	Sales
	Karan	IT
	Pooja	HR
	Raj	Sales
	Meena	IT
	Vikas	HR

Answer 23:

```
SELECT e.emp_name
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id
WHERE d.dept_name = 'IT';
```

	emp_name
►	Amit
	Karan
	Meena

Answer 24:

```
SELECT p.project_name, d.dept_name  
FROM Projects p  
JOIN Departments d ON p.dept_id = d.dept_id;
```

	emp_name	project_name	budget
►	Anuj	Alpha	100000
	Ravi	Delta	120000
	Ravi	Beta	150000
	Neha	Alpha	100000
	Amit	Gamma	200000
	Sneha	Delta	120000
	Sneha	Beta	150000
	Karan	Gamma	200000
	Pooja	Alpha	100000
	Raj	Delta	120000
	Raj	Beta	150000
	Meena	Gamma	200000
	Vikas	Alpha	100000

Answer 25:

```
SELECT e.emp_name, p.project_name, p.budget  
FROM Employees e  
JOIN Projects p ON e.dept_id = p.dept_id;
```

	emp_name	project_name	budget
►	Anuj	Alpha	100000
	Ravi	Delta	120000
	Ravi	Beta	150000
	Neha	Alpha	100000
	Amit	Gamma	200000
	Sneha	Delta	120000
	Sneha	Beta	150000
	Karan	Gamma	200000
	Pooja	Alpha	100000
	Raj	Delta	120000
	Raj	Beta	150000
	Meena	Gamma	200000
	Vikas	Alpha	100000

Answer 26:

```
SELECT MAX(salary) FROM Employees  
WHERE salary < (SELECT MAX(salary) FROM Employees);
```

	MAX(salary)
▶	70000

Answer 27:

```
SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) as salary_rank FROM  
Employees;
```

	emp_name	salary	salary_rank
▶	Karan	72000	1
	Amit	70000	2
	Meena	68000	3
	Sneha	65000	4
	Ravi	60000	5
	Neha	55000	6
	Raj	52000	7
	Vikas	51000	8
	Anuj	50000	9
	Pooja	48000	10

Answer 28:

```
SELECT * FROM Employees  
WHERE salary > (SELECT AVG(salary) FROM Employees);
```

	emp_id	emp_name	dept_id	salary	hire_date	city
▶	102	Ravi	2	60000	2021-03-15	Mumbai
	104	Amit	3	70000	2019-11-01	Pune
	105	Sneha	2	65000	2023-05-12	Mumbai
	106	Karan	3	72000	2018-08-25	Bangalore
	109	Meena	3	68000	2020-02-14	Pune
✱	NULL	NULL	NULL	NULL	NULL	NULL

Answer 29:

```
SELECT dept_id, SUM(salary) AS total_sal  
FROM Employees  
GROUP BY dept_id  
ORDER BY total_sal DESC  
LIMIT 1;
```

	dept_id	total_sal
▶	3	210000

Answer 30:

```
SELECT e.emp_name
FROM Employees e
JOIN Attendance a ON e.emp_id = a.emp_id
WHERE a.date = '2024-03-01' AND a.status = 'Absent';
```

	emp_name
▶	Ravi
	Sneha
	Meena